Trello Clone - Phase Breakdown & Roadmap

Project: Trello Clone Task Management Application
Current Phase: Phase 1 ✅ COMPLETE
Next Phase: Phase 2 (Planning)

Overview

This document outlines the development phases for the Trello Clone project, detailing what was accomplished in each phase and what's planned for future phases.

Phase 1: Core Foundation & MVP ✅ COMPLETED

Duration: [February 2026 - April 2026]
Status: ✅ Complete
Lead: Pratham, Prisha Chaudhary

Phase 1 Objectives

Build a functional, user-friendly task management application with core Trello-like features.

Phase 1 Features Delivered

1.1 User Authentication System

- ✅ **Sign Up**
  - Email validation
  - Password validation
  - User registration with Firebase Authentication
  - Form validation (empty field checks, email format)
- ✅ **Login**
  - Email & password authentication
  - Session token management
  - Remember user functionality
  - Error handling for invalid credentials
- ✅ **Logout**
  - Secure session termination
  - Clear local storage
  - Redirect to login page
- ✅ **Session Management**

    - Local storage for user session
    - Login persistence across browser refreshes
    - Auto-logout on expiration (future phase)

1.2 Task Management System

- ✅ **Create Tasks**
    - Add new tasks with title & description
    - Set priority levels (low, medium, high)
    - Assign due dates
    - Real-time Firebase integration
- ✅ **View Tasks**
    - Display all user tasks on dashboard
    - Filter by status (Pending, Running, Completed)
    - Task card UI with relevant information
    - Show task metadata (created date, assignee)
- ✅ **Update Tasks**
    - Edit task title & description
    - Change task priority
    - Modify due dates
    - Update task status
    - Real-time Firebase updates
- ✅ **Delete Tasks**
    - Remove tasks with confirmation
    - Firestore record deletion
    - UI update after deletion

1.3 Task Organization

- ✅ **Status Management**
    - Pending (To-Do)
    - Running (In Progress)
    - Completed (Done)
    - Visual status indicators on cards
- ✅ **Drag & Drop Interface**
    - HTML5 Drag and Drop API implementation
    - Drag tasks between status columns
    - Visual feedback during drag operations
    - Drop zone highlighting
    - Automatic status update on drop

- ✅ **Task Assignment**
  - Assign tasks to team members
  - Display assignee on task card
  - Assignee filtering (future phase)

1.4 User Interface

- ✅ **Responsive Design**
  - Desktop layout (primary)
  - Mobile layout (basic)
  - Flexible grid system
  - Touch-friendly buttons & interactions
- ✅ **Visual Design**
  - Clean, modern aesthetic
  - Consistent color scheme
  - Icon integration
  - Hover effects & transitions
  - Card-based UI layout
- ✅ **User Experience**
  - Intuitive navigation
  - Clear call-to-action buttons
  - Form validation feedback
  - Success/error messages
  - Loading states (basic)

1.5 Data Persistence

- ✅ **Firebase Integration**
  - Firebase Authentication setup
  - Firestore database for data storage
  - Real-time data synchronization
  - User-specific data isolation
- ✅ **Data Models**
  - User collection with profile data
  - Tasks collection with relationships
  - Proper data structure & indexing
  - Timestamp tracking (created, updated)

Phase 1 Architecture

Technology Used

```
Frontend:
├── HTML5 (Semantic markup)
├── CSS3 (Responsive design, Flexbox, Grid)
├── Vanilla JavaScript (ES6+)
└── HTML5 Drag & Drop API

Backend:
├── Firebase Authentication
├── Cloud Firestore
└── Firebase Hosting
```

Key Implementations

1. **Authentication Module**
    - Email/password based authentication
    - Firebase SDK integration
    - Session token management
    - Credential validation
2. **Task Management Module**
    - CRUD operations with Firestore
    - Status state machine
    - Real-time data binding
    - User-specific filtering
3. **Drag & Drop Module**
    - Drag event handlers
    - Drop zone management
    - Visual feedback system
    - Status update logic
4. **UI Module**
    - Responsive layout system
    - Component rendering
    - Event delegation
    - Dynamic DOM manipulation

Phase 1 File Structure

```
public/
├── index.html                 # Login/Signup page
├── dashboard.html            # Main dashboard
├── 404.html                  # Error page
├── styles/
│   ├── auth.css             # Authentication styling
```

```
│   └── dashboard.css        # Dashboard styling
└── scripts/
    ├── auth.js              # Authentication logic
    ├── dashboard.js         # Task management logic
    └── firebase-config.js   # Firebase config

Root:
├── firebase.json            # Firebase hosting config
├── .firebaserc             # Firebase project config
├── .gitignore              # Git ignore rules
└── Readme.md               # Project overview
```

Phase 1 Commits

1. Initial project setup
2. User authentication implementation
3. Task CRUD functionality
4. Drag-and-drop integration
5. Task assignment features
6. UI/UX updates & styling
7. Logout functionality

Phase 1 Testing

- ✅ Manual testing of all features
- ✅ Cross-browser compatibility (Chrome, Firefox, Edge)
- ✅ Mobile responsiveness testing
- ✅ Firebase integration verification
- ✅ Edge case handling (empty inputs, invalid data)

Phase 1 Known Limitations

1. No automated testing suite
2. No error logging system
3. Limited offline functionality
4. No real-time collaboration
5. Basic mobile experience
6. No performance optimizations
7. No analytics or monitoring

Phase 1 Lessons Learned

- Firebase Firestore provides excellent real-time capabilities
- Drag & Drop API is straightforward for task management
- Client-side validation is essential before Firebase calls
- CSS Grid/Flexbox provide good responsive foundations
- User experience design is critical for task management apps

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Phase 2: Enhanced Features & Collaboration ➡ SOON PLANNED

Estimated Duration: [May 2026 - July 2026]
Status: 📋 In Backlog
Objectives: Add collaborative features and improve user experience

Phase 2 Features (Planned)

2.1 Multi-Board Support

- [ ] Create multiple task boards per user
- [ ] Board management (create, edit, delete, archive)
- [ ] Board templates for quick setup
- [ ] Board-level settings & configuration
- [ ] Switch between boards easily

2.2 Collaboration Features

- [ ] Share boards with team members
- [ ] User roles (Owner, Editor, Viewer)
- [ ] Invite users via email
- [ ] Real-time presence indicators (who's viewing)
- [ ] User avatars & profile pictures
- [ ] Activity stream showing who did what

2.3 Comments & Activity

- [ ] Add comments to tasks
- [ ] Task activity log (history of changes)
- [ ] @mentions for notifications
- [ ] Email notifications (comment, assignment)
- [ ] Task timeline view

2.4 Advanced Task Features

- [ ] Task checklists (subtasks)
- [ ] File attachments/uploads
- [ ] Task labels & custom colors
- [ ] Recurring tasks
- [ ] Task dependencies
- [ ] Custom fields for tasks
- [ ] Task templates

2.5 Improved Search & Filtering

- [ ] Search tasks by keyword
- [ ] Advanced filter by status, priority, assignee
- [ ] Saved filters/views

- [ ] Sort tasks (by date, priority, name)
- [ ] Task archiving feature

2.6 Performance & Optimization

- [ ] Implement pagination for large task lists
- [ ] Virtual scrolling for efficiency
- [ ] Caching strategies
- [ ] Optimize Firebase queries
- [ ] Lazy loading for images
- [ ] Code splitting

Phase 2 Technical Improvements

- [ ] Unit testing framework setup (Jest)
- [ ] Integration testing
- [ ] Error boundary implementation
- [ ] Proper logging system
- [ ] Performance monitoring
- [ ] Better error messages

Phase 3: Mobile & Progressive Web App ➡ SOON PLANNED

Estimated Duration: [August 2026 - October 2026]
Status: 📋 In Backlog

Phase 3 Features

3.1 Progressive Web App (PWA)

- [ ] Service Worker implementation
- [ ] Offline functionality
- [ ] App icon & manifest
- [ ] Installable on home screen
- [ ] push notifications

3.2 Mobile-First Redesign

- [ ] Mobile optimization
- [ ] Touch gesture support
- [ ] Mobile navigation patterns
- [ ] Swipe to delete/archive
- [ ] Responsive improvements

3.3 Native Mobile Apps

- [ ] React Native / Flutter app
- [ ] iOS app deployment

- [ ] Android app deployment
- [ ] Push notifications
- [ ] Mobile-specific features

3.4 Offline Support

- [ ] IndexedDB for local data storage
- [ ] Sync queue for offline changes
- [ ] Conflict resolution
- [ ] Offline indicators

Phase 4: Enterprise Features & Scaling SOON PLANNED

Estimated Duration: [November 2026 - January 2027]
Status: 📋 In Backlog

Phase 4 Features

4.1 Team Management

- [ ] Organization/Workspace concept
- [ ] User roles & permissions (RBAC)
- [ ] Team settings & billing
- [ ] Member management dashboard
- [ ] Permission templates

4.2 Advanced Security

- [ ] Two-factor authentication (2FA)
- [ ] SSO integration (OAuth, SAML)
- [ ] API keys for integrations
- [ ] Audit logs for compliance
- [ ] Data encryption at rest
- [ ] IP whitelisting (enterprise)

4.3 Integrations & API

- [ ] REST API for external apps
- [ ] Webhook support
- [ ] Third-party integrations (Slack, Teams)
- [ ] Calendar integration
- [ ] Email integration
- [ ] Zapier/IFTTT support

4.4 Analytics & Reporting

- [ ] Dashboard analytics
- [ ] Team productivity metrics

- [ ] Time tracking per task
- [ ] Custom reports
- [ ] Export functionality (PDF, CSV)
- [ ] Usage analytics

Phase 5: Backend Migration & Infrastructure ➡ SOON PLANNED

Estimated Duration: [February 2027 - April 2027]
Status: 📋 In Backlog

Phase 5 Features

5.1 Backend Development

- [ ] Node.js/Express API server
- [ ] Database migration to PostgreSQL
- [ ] Schema design & migrations
- [ ] API documentation (OpenAPI/Swagger)
- [ ] Authentication refresh logic
- [ ] Rate limiting & throttling

5.2 DevOps & Infrastructure

- [ ] Docker containerization
- [ ] Kubernetes orchestration
- [ ] CI/CD pipeline (GitHub Actions)
- [ ] Automated testing (Unit, Integration, E2E)
- [ ] Performance monitoring (New Relic, Datadog)
- [ ] Log aggregation (ELK, Splunk)

5.3 Testing & Quality

- [ ] 80%+ code coverage
- [ ] Automated E2E tests (Cypress, Selenium)
- [ ] Load testing
- [ ] Security testing (OWASP)
- [ ] Performance profiling

5.4 Monitoring & Observability

- [ ] Application performance monitoring
- [ ] Error tracking (Sentry)
- [ ] Uptime monitoring
- [ ] Real-time alerting
- [ ] Health check endpoints

Phase 6: Advanced Features & Optimization ➡ SOON PLANNED

Estimated Duration: [May 2027 onwards]
Status: 📋 In Backlog

Phase 6 Features

- [ ] AI-powered task recommendations
- [ ] Natural language processing for task creation
- [ ] Automated task scheduling
- [ ] Learning algorithms for user behavior
- [ ] Advanced analytics & predictive insights
- [ ] Mobile app v2 with native features
- [ ] Desktop app (Electron)
- [ ] Browser extensions
- [ ] Voice commands integration

Roadmap Summary

Timeline

```
Phase 1 ✅  [Feb 2026 - Apr 2026] - Core MVP
    ↓
Phase 2 ➡️  [May 2026 - Jul 2026] - Collaboration
    ↓
Phase 3 ➡️  [Aug 2026 - Oct 2026] - Mobile & PWA
    ↓
Phase 4 ➡️  [Nov 2026 - Jan 2027] - Enterprise
    ↓
Phase 5 ➡️  [Feb 2027 - Apr 2027] - Backend Migration
    ↓
Phase 6 ➡️  [May 2027 - ] - Advanced Features
```

Feature Distribution by Phase

Category	Phase 1	Phase 2	Phase 3	Phase 4	Phase 5	Phase 6
Core Task Management	✅	📈	📈	-	-	-
Authentication	✅	📈	-	📈	-	-
Collaboration	-	✅	-	📈	-	-
Mobile	✅	-	✅	-	-	📈
Enterprise	-	-	-	✅	📈	-
Infrastructure	🔧	🔧	🔧	🔧	✅	📈

Legend: ✅ Full, 📈 Enhanced, 🔧 Basic, 📋 Planned

Success Metrics

Phase 1 Metrics (Achieved)

- ✅ User authentication working
- ✅ CRUD operations functional
- ✅ Drag-drop interface operational
- ✅ Firebase integration complete
- ✅ Responsive design working
- ✅ All core features implemented

Future Phase Metrics

- User engagement & retention
- Feature adoption rates
- Performance metrics (load time, response time)
- User satisfaction (NPS score)
- Security audit results
- Scalability benchmarks

Dependencies & Risks

Phase 1 Risks (Mitigated)

- ✅ Firebase learning curve - Resolved
- ✅ Drag & Drop browser compatibility - Verified
- ✅ Cross-browser testing - Completed

Phase 2 Risks

- Real-time collaboration complexity
- Firestore scalability limits
- Managing user permissions

Phase 3 Risks

- Mobile platform differences
- App store approval process
- Offline sync conflicts

Phase 4+ Risks

- SSO integration complexity
- Regulatory compliance (GDPR, CCPA)
- Enterprise security requirements
- High infrastructure costs

Resource Allocation

Phase 1 (Completed)

- **Frontend Developer:** 1 FTE
- **Backend Developer:** 0.5 FTE (Firebase)
- **QA/Testing:** 0.5 FTE
- **Total:** 2 people

Phase 2 (Estimated)

- **Frontend Developer:** 1.5 FTE
- **Backend Developer:** 1 FTE
- **QA/Testing:** 1 FTE
- **Total:** 3.5 people

Phase 3 (Estimated)

- **Frontend Developer:** 1 FTE
- **Mobile Developer:** 2 FTE (iOS, Android)
- **Backend Developer:** 1 FTE
- **DevOps:** 0.5 FTE
- **QA/Testing:** 1.5 FTE
- **Total:** 6 people

Communication & Stakeholders

Current Team

- **Pratham** - Full Stack Developer
- **Prisha Chaudhary** - Frontend Developer

Stakeholders

- Product Owner / Project Manager
- QA Team
- Security Team (future phases)
- DevOps Team (Phase 5+)

Meeting Schedule

- Weekly sprint kicks-off (Product discussion)
- Bi-weekly progress reviews
- Ad-hoc technical discussions

Conclusion

Phase 1 of the Trello Clone project is complete with all core features delivered. The foundation is solid for expanding into collaborative features, mobile support, and enterprise-grade functionality in subsequent phases.

The project demonstrates:

- ✅ Effective use of modern web technologies
- ✅ Clean, maintainable code architecture
- ✅ User-focused design
- ✅ Scalable backend infrastructure (Firebase)

The roadmap provides a clear path for evolution while maintaining quality and user experience.

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